Resilience & Recovery — Jahna Deployment

Date: 2026-05-27 · **For:** Ops + Sales budget meeting (Dividia internal)
Purpose: ground the "lightning-grade reliability" requirement in concrete failure modes, define what Standard vs Failover tier actually includes, and lay out a non-technical equipment-swap playbook so the scale operator can keep loads moving when something breaks.

> **The thesis.** Jahna's site gets hit by lightning several times per year. A single bad strike can take out cameras, switches, the scale itself, or any combination — and the load-out operation has to keep moving. We design for two outcomes: **(1) when a single piece of equipment fails, the rest of the system keeps weighing and recording, and the operator gets a clear next-step**; and **(2) when a non-technical person swaps a broken box, they can do it in under five minutes without calling Dividia.** Both outcomes need to be in the quote — they are not free, but they are the difference between "we ship and forget" and "we ship and own the support burden."

1. Failure mode taxonomy — what can break

Every device with a wire to the outside world is a lightning target. Every device with a network connection is a software-failure target. Every device with moving parts (HDD, fan) is a wear target. Listing in severity-to-operations order:

What fails	Severity	Operations impact	Auto-recovery	Manual playbook
Scale (down, others up)	CRITICAL	No weight, no scale-on event. Transaction cannot complete.	None possible. Scale is essential equipment.	Operator-fallback flow (see § 3). Trucks weighed on outbound only or sent back.
NVR	CRITICAL	No transaction recording. Cameras still alive but un-recorded.	systemd restart, watchdog. If hardware: none.	Cold-spare swap (~5 min) on Failover tier.
Jetson Spot Decoder	HIGH	No measurement. Operator sees blank display.	systemd restart of spot-decoder process.	Hot-spare swap (~3 min) on Failover. Trucks process with `Alert_NoMeasurement`; print-time snapshot still captured.
QR permit camera	HIGH	No automatic permit read.	RTSP exponential reconnect (1 → 60 s).	PoE power-cycle from switch; manual permit-number entry via spot decoder UI; print-time QR snapshot saved as evidence.
Measurement camera	HIGH	No side-view, operator can't measure.	RTSP reconnect.	PoE power-cycle; fall back to `Alert_NoMeasurement` + print-time evidence frame.
Internet / WAN	MEDIUM	Cloud portal unreachable, FastWeigh webhooks delayed.	Recorder queues uploads locally and flushes on reconnect.	Check router; LTE failover on Failover tier.
FastWeigh cloud / cloudapi	MEDIUM	Webhooks not arriving even with local internet up.	cloudapi watchdog restarts; recorder queues locally.	Reconciliation worker (Failover tier) sweeps missed tickets via REST/GraphQL.
Power (whole site)	HIGH	Site stops. UPS gives graceful shutdown.	UPS → graceful shutdown of NVR + Jetson with INI flush.	Site generator if available; ISP call for outage.
Lightning strike (multi-device)	CRITICAL	Could be one box or all of them. Random scatter.	Surge protection clamps the surge before it reaches devices (if installed correctly).	On-site kit swap of damaged devices; SMS alert kicks in; Dividia ops mobilizes within 1 hour.
HDMI display	LOW	Operator screen blank; spot decoder still running.	None.	Swap HDMI cable or display; spot decoder unaffected.
Operator input controller (Razer Stream Controller)	LOW	Operator can't mark lines with the knobs.	None.	Keyboard fallback (built into spot decoder); swap controller from on-site kit. Unit is the same Razer Stream Controller Jahna already uses with AxlEye today.

2. The "scale down, everything else up" scenario — load-bearing case

Specifically called out because this scenario is the difference between a system that *kind of* works and one that survives bad days. If the scale dies but our cameras + Jetson + NVR are still alive, we have a choice: stop the operation entirely, or capture as much as we can and reconcile later.

> **Design call.** The transaction trigger should NOT be exclusively scale-on. Scale-on is the *primary* trigger — it is the most accurate signal that a truck is here. But it must have a secondary trigger so a scale failure doesn't paralyze the rest of the pipeline. The two viable secondary triggers are: **(a) QR-scan event** (driver shows permit → transaction opens) and **(b) operator manual-start button** on the spot decoder display. Both should be in the design.

What happens when the scale dies:

1. Scale heartbeat times out (recorder logs `Scale_HeartbeatLost`).
2. Banner appears on the spot decoder + scale-house dview: **"SCALE OFFLINE — transactions in manual mode"**. SMS alert fires to Dividia ops + Jahna technical contact.
3. Operator can still: read permit (QR cam alive), measure axles (Jetson alive), capture truck images (cameras alive).
4. Operator decides per truck: (a) send back, (b) record permit + measurement + photos with no weight and reconcile on next visit, or (c) weigh on a different scale at the same site if one exists (Jahna has 1–2 inbound + 1–2 outbound per site).
5. Transactions tagged `Scale_Offline` in the cloud audit trail so they're searchable later.
6. When scale comes back, normal flow resumes. Any pending manual transactions stay in the portal as "needs weight reconciliation."

Quote line item: ~12–20 engineering hours to add the secondary triggers, fallback UI, alert path, and reconciliation tagging. Included in Phase 1 base — this is not optional.

3. Per-failure deep dive

Each scenario follows the same shape: **symptom** (what operator sees), **detection** (how we know automatically), **auto-recovery** (what fixes itself), **manual intervention** (what the operator does, with no Dividia call needed), **permanent fix** (what Dividia does later).

3.1 NVR down

Symptom	dview can't reach NVR; spot decoder loses its event sink; cloud portal shows site as offline.
Detection	cloudapi WebSocket disconnect within 5 s; site-health dashboard turns red.
Auto-recovery	systemd restart of recorder + connector services. If hardware failure: none.
Manual intervention	Power-cycle the NVR once (back-of-rack switch labeled "NVR POWER"). If not back in 60 s, see the spare-swap playbook in § 4. Spare swap target: ≤5 min.
Permanent fix	Failed unit shipped to Dividia for RMA; new spare ships the same day.

3.2 Jetson Spot Decoder down

Symptom	Operator's measurement display goes black. HTTP API on :88 not responding.
Detection	recorder's measurement_client times out; spot-decoder heartbeat lost.
Auto-recovery	systemd restart of spot-decoder process (typical recovery <10 s).
Manual intervention	Power-cycle Jetson via labeled switch. If not back, swap from hot spare (Failover tier). Operator can keep processing trucks — transactions log `Alert_NoMeasurement` and continue with permit + ticket + scale weight. Print-time fallback (already shipped on the recorder branch) snags the latest side-view frame as evidence even without a saved measurement.
Permanent fix	Failed unit returned for diagnosis; Dividia images a fresh Jetson and overnights it.

3.3 QR permit camera down

Symptom	QR camera live view absent on dview; `Qr_Read` events stop firing.
Detection	ThreadQrReader logs RTSP reconnect attempts (exponential 1 → 60 s).
Auto-recovery	RTSP reconnect loop. If camera itself froze, PoE power-cycle from the network switch typically clears it (switches that support per-port PoE toggle are part of the BOM).
Manual intervention	Operator triggers PoE power-cycle from a button on the dview camera setup screen (one-click). If still down, operator can enter permit number manually in the spot decoder UI; recorder calls FDOT PAS directly with the typed GUID. Transaction completes with `Permit_ManualEntry` flag for audit.
Permanent fix	Spare Uniview camera in on-site kit; pre-configured DHCP + RTSP credentials so swap is plug-and-play.

3.4 Measurement camera down

Symptom	Side-view feed on spot decoder display goes black or freezes.
Detection	Spot decoder logs FFmpeg input loss; recorder gets stale measurements.
Auto-recovery	FFmpeg reconnect on spot decoder. RTSP retry on recorder side.
Manual intervention	PoE power-cycle via dview; if dead, swap from on-site kit. Trucks still process with `Alert_NoMeasurement`.
Permanent fix	Same as QR camera — spare-swap path.

3.5 Internet / WAN down

Symptom	Cloud portal unreachable, FastWeigh webhooks not arriving.
Detection	cloudapi reachability probe; site-health dashboard yellow within 30 s.
Auto-recovery	Recorder queues transaction uploads to local disk (already implemented — `thread_cloud_sw` retries until 200, deletes on success). FastWeigh webhooks queue at TAC Insight per their Svix retry policy (one of the things to confirm with Brad).
Manual intervention	Check router lights; call ISP if total outage. **Failover tier:** LTE backup uplink (Cradlepoint or similar) kicks in automatically within ~60 s and traffic shifts. Cost: ~$60/mo/site for the LTE plan, hardware $300–500.
Permanent fix	ISP service ticket. Failover tier prevents this from being operationally visible at all.

3.6 FastWeigh cloud or Dividia cloudapi down

Symptom	Our local internet is up but webhooks aren't arriving / cloudapi returns 5xx.
Detection	Heartbeat from cloudapi missing; webhook freshness gauge flat-lines.
Auto-recovery	cloudapi auto-restart. NVR queues locally.
Manual intervention	None at the site — this is a Dividia ops responsibility. Operator just sees "cloud temporarily offline; transactions queued."
Permanent fix	**Failover tier:** reconciliation worker polls FastWeigh REST/GraphQL for "tickets since X" once cloudapi is back, sweeps anything that didn't reach us during the outage. This is the line item that depends on Brad's webhook-replay answer.

3.7 Lightning strike — the multi-device scenario

Symptom	Could be anything. Site partially or fully down. Operator may smell ozone or see fault LEDs.
Detection	Multiple heartbeats drop near-simultaneously; SMS alert fires.
Auto-recovery	**Prevention is everything.** Surge protective devices (SPDs) on every wire entering the scale house — power, Ethernet, scale serial. Proper grounding back to a single point. Fiber media converters on outdoor camera runs where feasible (fiber is not a lightning conductor; copper Ethernet is).
Manual intervention	Power-cycle everything from the rack PDU. Anything that doesn't come back gets swapped from on-site kit using the playbook in § 4. SMS alert ensures Dividia ops is engaged within minutes; we coordinate replacement-unit ship from Dividia same day.
Permanent fix	Damaged units RMA'd to Dividia. SPD inspection + replacement on the lines that conducted. Periodic site audit (annual) of surge protection state.

4. Non-technical equipment swap playbook

The goal: a scale operator with no IT background should be able to swap a failed Jetson, NVR, camera, or controller in under five minutes, and know they did it right before they walk away. Six design principles get us there.

4.1 Label everything in plain English

Every cable end carries a printed tag at both ends: "**MEASUREMENT CAMERA → JETSON PORT 1**", "**QR CAMERA → POE SWITCH PORT 3**", "**NVR → ROUTER**". No part numbers, no jargon. The operator should never have to guess what plugs into what.

4.2 Color-code by function

Network cables: **green = measurement camera**, **blue = QR camera**, **yellow = scale data**, **white = internet uplink**, **black = power**. Colored boots / wraps on existing cables — cheap, retrofittable. The operator can match by color in 2 seconds.

4.3 Pre-imaged recovery media on every device

Each Jetson has a labeled USB stick fastened to the underside of its case: "**RESTORE THIS JETSON — boot from this USB if device won't start.**" The image carries the site-specific config (RTSP URLs, calibration, network settings). Same pattern for the NVR.

4.4 "This is broken" workflow in the scale-house UI

A page in dview titled **"Something is broken"**. Operator picks from a dropdown: "My measurement display is dead", "My QR camera isn't reading", "The scale isn't connected", etc. Selecting a row immediately:

- Prints a one-page runbook tailored to that failure (the playbook from § 3).
- Sends an alert to Dividia ops + Jahna primary contact.
- Logs the failure in the site's history for the next maintenance visit.
- Walks the operator through 3 verification questions after the swap so they know it worked.

This removes thinking from the equation. The operator never has to remember "what was that troubleshooting step?" — they pick the problem and follow the printed sheet.

4.5 Five-minute swap target

Equipment chosen and mounted so swap is plug-out, plug-in — no screwdrivers, no boot menus, no IP config. Jetson on quick-release DIN rail; NVR on slide rails; cameras on quick-release mounts. The on-site kit lives in a labeled drawer near the scale house.

4.6 Post-swap confidence check

After the swap, the dview UI walks the operator through three checks:

1. "Do you see the green LED on the new device?" → if no, check power cable.
2. "Can you see the side-view camera feed on this screen?" → if no, check camera cable.
3. "Press the joystick. Does a line appear on the screen?" → if yes, you're done.

Three questions, three known-good outcomes. Operator walks away with confidence the swap worked.

5. On-site spare kit (per site)

The minimum kit lives in a labeled toolbox at the scale house. Contents:

- 1 × pre-imaged Jetson (with USB recovery + site config baked in)
- 1 × pre-imaged NVR (cold spare, smaller form-factor than primary)

- 1 × Razer Stream Controller (w/ knobs) — matches operator's existing AxlEye-era device; ~$270 list, ~$120 used like-new
- 1 × HDMI cable (10 ft) and 1 × HDMI display (small, ~10")
- 1 × Uniview camera (matched model, pre-configured)
- 2 × CAT6 PoE patch cables (color-coded blue, green) at 10 ft + 25 ft
- 2 × outlet-grade surge protectors
- 1 × labeled USB recovery stick for each device type
- Printed quick-reference card: "What broke? Look on the front of this kit."

Kit cost per site: roughly $1,500–2,200 depending on Jetson + camera spare choices. Failover tier sites include the kit; Standard tier sites can buy in optionally.

6. Heartbeat & monitoring

Every device pings cloudapi every 30 s with its identity + uptime + temperature (where available). Aggregation gives us per-site green / yellow / red on a dashboard. Alert escalation:

Tier	What's monitored	Alert channel	Response SLA
Standard	NVR, Jetson, both cameras, internet uplink	Email (Dividia ops + Jahna contact)	Next business day
Failover	Above + UPS battery state + scale heartbeat + cloudapi reachability + FastWeigh webhook freshness	SMS + email; PagerDuty for Dividia ops	1-hour engagement, 4-hour onsite escalation if needed

7. Reliability tier definition (concrete scope)

Standard tier — ~40–80 hr eng + $500–800/site hardware

- UPS sized for graceful shutdown on NVR + Jetson + scale-house network switch (15–30 min runtime)
- Surge protection on power + on Ethernet entering scale house + on scale serial line
- Heartbeat monitoring + email alerts
- Pre-imaged USB recovery media on every primary device
- Labeled cables + color-code retrofit on existing runs
- Printed scale-house runbook (laminated, mounted by rack)
- "Something is broken" workflow in dview UI

Failover tier — additional ~80–120 hr eng + $300–500/site hardware (above Standard)

- Hot-spare Jetson on-site (pre-configured, plug-and-play swap)
- Cold-spare NVR on-site (matched config, pre-imaged USB)
- LTE backup uplink (Cradlepoint or similar) with automatic failover
- Reconciliation worker in cloudapi — polls FastWeigh REST/GraphQL for tickets missed during outages (scoped contingent on Brad's webhook-replay answer)
- SMS + PagerDuty alert escalation
- Site-health dashboard visible to Dividia ops + Jahna primary contact
- On-site spare kit included
- Annual site visit for surge / grounding / UPS audit

8. Site-walk checklist (Van Fleet first visit)

Before any equipment lands at Van Fleet, this gets done. Sets up everything above for success.

- **Grounding inspection** — is the scale house tied to a single grounding point? What's the resistance to true ground? Anything floating?
- **Surge protection inventory** — what's currently in place at the service entry, the scale serial line, the camera Ethernet runs?
- **UPS inventory + sizing** — what's running on what UPS? Battery age? Runtime under current load?
- **Outdoor cable run audit** — every meter of copper Ethernet outside the building is a lightning antenna. Where can we move to fiber? Where is fiber overkill?
- **Network topology map** — switches, PoE budgets, runs, room for growth.
- **Spare parts storage** — where does the on-site kit physically live? Is it dry? Lockable?
- **Generator + transfer switch status** — does the site have backup power? Does the scale house make it onto that circuit?
- **Cellular signal survey** — for Failover tier LTE backup; verify Verizon + AT&T coverage at the building.
- **Photo + measure** every camera mount, rack location, cable run, and outlet.

Output: a one-page site spec we lock down before procurement. Whatever surge / grounding / UPS gaps we find feed back into the customer quote as line items.

Cross-references: model-shift memory 2026-05-26 (lightning-grade requirement, Jetson replaces AxlEye PC, scale-on trigger), open-questions.md §1.0b (print-time fallback already implemented), worklog.md (~107–130 hr sunk).